

- 1) **A configuration service must expose a fixed list of supported regions that client code must not modify. Propose a modern collection-creation strategy and explain the difference between an unmodifiable view and a truly immutable collection.**
[Recent Collection Practices]

Aim

To create a fixed list of supported regions using Java's modern immutable collection (List.of()) and understand the difference between an unmodifiable view and a truly immutable collection.

Pseudocode

- Start
- Create a normal list
- Add "Asia", "Europe" to the list
- Create an unmodifiable view of the list
- Modify the original list by adding "America"
- Display the unmodifiable view
// Output includes "America" because the original list changed
- Create an immutable list using List.of("Asia", "Europe")
- Attempt to add "America"
- Display "UnsupportedOperationException"
- End

CODE

```
import java.util.ArrayList;
import java.util.Collections;
import java.util.List;

public class Main {

    public static void main(String[] args) {

        // Unmodifiable View
        ArrayList<String> regions = new ArrayList<>();
        regions.add("Asia");
        regions.add("Europe");
        List<String> view = Collections.unmodifiableList(regions);
        System.out.println("Unmodifiable View: " + view);

        // Modify the original list
        regions.add("America");
```

```

System.out.println("After modifying original list:");
System.out.println("Unmodifiable View: " + view);
// Truly Immutable Collection
List<String> immutableRegions = List.of("Asia", "Europe", "America");
System.out.println("\nImmutable Collection: " + immutableRegions);
// Attempt to modify immutable collection
try {
    immutableRegions.add("Africa");
} catch (UnsupportedOperationException e) {
    System.out.println("Cannot modify immutable collection.");
}
}

```

OUTPUT:

```

Unmodifiable View: [Asia, Europe]
After modifying original list:
Unmodifiable View: [Asia, Europe, America]

Immutable Collection: [Asia, Europe, America]
Cannot modify immutable collection.

```